

Experiment 1: Human Activity Recognition in Video using Pose Estimation

Aim:

To recognize human activities in a video stream by estimating body poses using the MediaPipe Pose model.

Theory:

Pose estimation is a technique in computer vision that detects and tracks key body joints such as shoulders, elbows, hips, and knees. MediaPipe Pose is a deep-learning-based model that identifies 33 human body landmarks from RGB images or video frames. This helps us recognize human actions like walking or waving by analyzing the motion and relative position of joints across frames. In real-time applications, the system continuously tracks and updates body poses to reflect ongoing movement patterns. When combined with object detection (e.g., YOLO), pose estimation can be applied specifically to regions containing humans, improving accuracy in multi-person scenes.

Algorithm (Version A: MediaPipe Only):

- Start
- Import required libraries (cv2, mediapipe)
- Load the input video using cv2.VideoCapture()
- Initialize MediaPipe Pose model and drawing utility
- Loop through each video frame:
 - - Read frame from video
 - - Convert frame from BGR to RGB
 - - Use pose.process() to detect landmarks
 - - Draw pose landmarks and connections
- Display frame with pose overlay
- Break loop on 'q' key press or end of video
- Release video and destroy all OpenCV windows
- End

Algorithm (Version B: YOLO + MediaPipe for Multi-Person Detection):

- Start
- Import cv2, mediapipe, and ultralytics.YOLO
- Load input video using cv2.VideoCapture()
- Load YOLOv8 model (yolov8n.pt) for detecting people
- Initialize MediaPipe Pose model
- For each video frame:
 - - Resize and convert to RGB
 - - Use YOLO to detect humans (class = 0)
 - - For each detected bounding box:
 - - Crop the person region
 - - Convert crop to RGB
 - - Apply pose detection using pose.process()
 - - Draw landmarks on cropped region

- - Replace cropped region back into full frame
- Display final frame with pose overlays
- Stop when 'q' is pressed
- Release resources and close all windows
- End

Flowchart (MediaPipe-Only Version):

Start → Import Libraries → Load Video → Initialize Pose Model → Read Frame → Convert BGR to RGB → Detect Pose → Draw Landmarks → Display Frame → Quit on Key Press → End

Viva Questions:

- What is the purpose of converting BGR to RGB in pose estimation?
- How many landmarks does MediaPipe Pose detect?
- What are the advantages of combining YOLO with MediaPipe?